T+ TECHPLUS

2026 Edition

MICROSERVICES

Complete Learning Roadmap

Split systems into independently deployable services with robust communication and observability.

- Service boundaries
- Domain-driven basics
- API contracts
- Service ownership
- Sync vs async communication
- Event versioning

01 Fundamentals

02 Core Concepts

03 Tools and Frameworks

04 Real Projects

05 Advanced Topics

5	**20+**	**10+**	**100%**
Learning Stages	Core Topics	Projects	Microservices

Learning Path Overview

Five progressive stages from fundamentals to production mastery

01
Fundamentals

Service boundaries • Domain-driven basics • API contracts • Service ownership

02
Core Concepts

Sync vs async communication • Event versioning • Data consistency approaches • Resilience patterns

03
Tools and Frameworks

API gateway • Service mesh • Broker technologies • Distributed tracing tools

04
Real Projects

Order workflow with saga • Inventory service split • Audit/event pipeline • Independent deploy pipeline

05
Advanced Topics

Cross-service governance • Platform APIs • Latency budgeting per service • Incident isolation strategy

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

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Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Service boundaries
- Domain-driven basics
- API contracts
- Service ownership

Tools & Frameworks

- Service boundaries
- Domain-driven basics
- API contracts
- Service ownership

Recommended Projects

- Microservices workshop using Service boundaries
- Microservices portfolio project with Domain-driven basics

Stage 1 Milestone Complete Microservices workshop using Service boundaries and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Sync vs async communication
- Event versioning
- Data consistency approaches
- Resilience patterns

Tools & Frameworks

- Sync vs async communication
- Event versioning
- Data consistency approaches
- Resilience patterns

Recommended Projects

- Microservices workshop using Sync vs async communication
- Microservices portfolio project with Event versioning

Stage 2 Milestone Complete Microservices workshop using Sync vs async communication and document the decisions, trade-offs, and validation checks before moving to Stage 3.

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Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- API gateway
- Service mesh
- Broker technologies
- Distributed tracing tools

Tools & Frameworks

- API gateway
- Service mesh
- Broker technologies
- Distributed tracing tools

Recommended Projects

- Microservices workshop using API gateway
- Microservices portfolio project with Service mesh

Stage 3 Milestone Complete Microservices workshop using API gateway and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Order workflow with saga
- Inventory service split
- Audit/event pipeline
- Independent deploy pipeline

Tools & Frameworks

- Order workflow with saga
- Inventory service split
- Audit/event pipeline
- Independent deploy pipeline

Recommended Projects

- Microservices workshop using Order workflow with saga
- Microservices portfolio project with Inventory service split

Stage 4 Milestone Complete Microservices workshop using Order workflow with saga and document the decisions, trade-offs, and validation checks before moving to Stage 5.

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Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

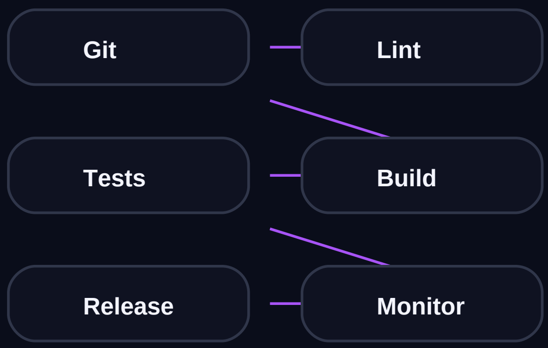
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Cross-service governance
- Platform APIs
- Latency budgeting per service
- Incident isolation strategy

CI/CD Pipeline



Stage 5 Milestone

A production-grade microservices workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

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Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE Microservices docs and community guides
Link: https://microservices.io/

FREE Build one portfolio project per stage before moving ahead
Link: https://roadmap.sh/software-architect

PAID Structured specialization track focused on Microservices
Link: https://roadmap.sh/software-architect

FREE Microservices roadmap reference
Link: https://roadmap.sh/software-architect

FREE Official documentation
Link: https://microservices.io/

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Microservices?
- Which project best proves Stage 4 competency in Microservices?
- What trade-offs appear in Stage 5 decisions for Microservices?